

Random circulant theta: an all-time fixed point and a root-moment criterion

Abstract

The random-circulant expectation problem MD-02 asks whether $\mathbb{E}\vartheta(G_n)/\sqrt{n} \rightarrow 1$ for the inversion-class Bernoulli ensemble, including even orders. We give an exact fixed-point representation of the complementary weighted logarithmic barriers. It is valid at every finite barrier time and comes with a deterministic global convergence bound for Picard iteration. Complement symmetry converts the target into an explicit moment condition on one coordinate of the converged fixed point. For the first two Picard iterates we prove uniform all-orders bounds $n\mathbb{E}(u_0^{(1)})^2 \leq 2$ and $n\mathbb{E}(u_0^{(2)})^2 \leq 68$, without an assumption on the barrier time. These are finite-iterate theorems, not bounds on the converged root. The required probabilistic control as the iteration count increases is not proved here. Consequently this manuscript does not prove or disprove MD-02.

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1 Target and scope

For $n \geq 2$, let $S = -S \subset \mathbb{Z}_n \setminus \{0\}$ be obtained by including each inversion class $\{s, -s\}$ independently with probability $1/2$. The graph G_S has an edge between distinct i, j exactly when $i - j \in S$. The class $\{n/2\}$ is sampled once when n is even. The displayed target in MD-02 is

$$\frac{\mathbb{E}\vartheta(G_S)}{\sqrt{n}} \longrightarrow 1 \quad \text{through all integer orders.} \quad (1)$$

The problem and the repository's status conventions are recorded in [1, 2]. The primary random-circulant theta paper is [3]. The earlier supplied manuscripts develop central-path skew identities, initial moments, and early-time estimates [4, 5]. Here we use a different representation of the same complementary barriers. All mathematical results used in the new reduction are proved below.

The scope distinction is essential. The fixed point exists and is uniquely specified for every deterministic mask, and its first two iterates have small second moments in the random ensemble. Neither fact proves that the converged coordinate has a small second moment. No passage from two iterations to an unbounded number of iterations is made implicitly.

2 The Fourier probability program

Let F be the unitary discrete Fourier transform, with $F_{sj} = n^{-1/2} \exp(-2\pi isj/n)$. We use the unnormalized transform $\hat{p} = \sqrt{n}Fp$ for probabilities. Averaging the standard theta semidefinite program over cyclic translations gives

$$\vartheta(G_S) = n \max\{p_0 : p \in \mathbb{R}_{\geq 0}^n, \mathbf{1}^T p = 1, \hat{p}_s = 0 \ (s \in S)\}. \quad (2)$$

For completeness, a translation-invariant primal matrix has entries $X_{ij} = f(i - j)/n$, with $f(0) = 1$ and $f(s) = 0$ on edges. Its positive semidefiniteness is equivalent to nonnegativity of the Fourier eigenvalues. Writing $p = \hat{f}/n$ gives a probability vector and objective $\sum_s f(s) = np_0$. Real symmetric matrices correspond to even p . Conversely, any real feasible probability can be averaged with its reflection; this preserves p_0 and the constraints. Thus allowing all real probabilities in (2) does not change its value. Complex equalities in that formula mean both their real and imaginary parts.

Encode the mask by $\sigma_0 = 0$, $\sigma_s = -1$ for edges and $\sigma_s = +1$ for nonedges, and define

$$R = F^* \text{diag}(\sigma) F, \quad P_0 = \frac{1}{n} \mathbf{1} \mathbf{1}^T. \quad (3)$$

Then R is a real symmetric circulant matrix and

$$R\mathbf{1} = 0, \quad R^2 = I - P_0, \quad \|R\|_{2 \rightarrow 2} = 1. \quad (4)$$

The complementary graph changes R to $-R$. Let $r = Re_0$ denote the first column. Because R is symmetric and circulant, $R_{0j} = r_j$.

Lemma 2.1 (Complementary feasible probabilities). *If p is feasible in (2) and q is feasible for the complementary graph, then $p^T q = 1/n$. In particular $np_0 \leq 1/q_0$ when $q_0 > 0$.*

Proof. The nonzero Fourier supports of p and q are disjoint. Parseval's identity therefore gives $p^T q = \hat{p}_0 \hat{q}_0 / n = 1/n$. Nonnegativity yields $p_0 q_0 \leq 1/n$. $\square$

3 Complementary weighted barriers

Fix $t \geq 1$, and set

$$w_0 = t, \quad w_j = 1 \ (j \neq 0), \quad W = t + n - 1, \quad v = w/t, \quad V = W/t = 1 + (n - 1)/t. \quad (5)$$

Maximize $\sum_j w_j \log p_j$ over the strictly positive feasible probabilities for G_S . The uniform probability is feasible. The objective is strictly concave and tends to minus infinity at the boundary, so it has a unique strictly positive maximizer $p = p(t)$.

Lemma 3.1 (Weighted complementary pair). *There is a unique positive complementary probability $q = q(t)$ satisfying*

$$p_j q_j = \frac{w_j}{nW} \quad (0 \leq j < n). \quad (6)$$

Moreover q is the weighted-barrier maximizer for the complementary graph.

Proof. The first-order optimality condition says that w/p , with division coordinatewise, belongs to the span of $\mathbf{1}$ and the real Fourier constraint vectors on S . Write $w/p = \lambda \mathbf{1} + h$, with h in the edge Fourier space. Taking the inner product with p gives $\lambda = W$. Define $q = (w/p)/(nW)$. It is positive, sums to one, and its Fourier support is contained in $S \cup \{0\}$. Thus it is complementary feasible and satisfies (6). Conversely $w/q = nWp$ gives its own first-order optimality condition. Strict concavity proves uniqueness on both sides. $\square$

Proposition 3.2 (Deterministic theta interval). *The weighted pair satisfies*

$$np_0 \leq \vartheta(G_S) \leq \frac{1}{q_0} = \frac{W}{t} np_0. \quad (7)$$

For every circulant graph, $\vartheta(G_S)\vartheta(\overline{G_S}) = n$.

Proof. The first inequality is primal feasibility and the second follows from Lemma 2.1; the equality uses (6). For any optimal complementary probabilities the same lemma gives the product upper bound n . The weighted feasible pair gives the product lower bound nt/W for every t . Letting $t \rightarrow \infty$ proves equality. $\square$

4 The all-time fixed point

All square roots in what follows are coordinatewise. Define

$$\Phi_{R,t}(u) = R\sqrt{v + u^2}. \quad (8)$$

Theorem 4.1 (Exact fixed-point representation). *For every mask and every finite $t \geq 1$, the equation*

$$u = R\sqrt{v + u^2} \quad (9)$$

has exactly one solution $u^ = u^*(R, t)$ in $\mathbb{R}^n$. Put $c = \sqrt{v + (u^*)^2}$. The complementary barrier pair is*

$$p = \frac{c + u^*}{\sqrt{nV}}, \quad q = \frac{c - u^*}{\sqrt{nV}}. \quad (10)$$

In particular,

$$\sqrt{\frac{t}{W}}(\sqrt{1 + (u_0^*)^2} + u_0^*) \leq \frac{\vartheta(G_S)}{\sqrt{n}} \leq \sqrt{\frac{W}{t}}(\sqrt{1 + (u_0^*)^2} + u_0^*). \quad (11)$$

Complementation changes u^* to $-u^*$.

Proof. Start with Lemma 3.1 and define $c = \sqrt{nV}(p+q)/2$, $u = \sqrt{nV}(p-q)/2$. Equation (6) gives $c_j^2 - u_j^2 = v_j$, and $c_j > 0$. At nonzero Fourier frequencies the two probabilities have complementary supports, so $u = Rc$. This proves existence.

Conversely suppose (9) holds, and put $c = \sqrt{v+u^2}$. By (4), $\mathbf{1}^T u = 0$ and

$$\|u\|^2 = \|c\|^2 - \frac{(\mathbf{1}^T c)^2}{n}.$$

Since $\|c\|^2 - \|u\|^2 = V$ and c is positive, $\mathbf{1}^T c = \sqrt{nV}$. Thus (10) defines positive probabilities. Their Fourier supports are complementary, and their coordinate products are $v_j/(nV) = w_j/(nW)$. They satisfy the first-order conditions of Lemma 3.1, so uniqueness of the barrier maximizer proves uniqueness of u .

At coordinate zero, $c_0 = \sqrt{1+u_0^2}$ and $np_0/\sqrt{n} = \sqrt{t/W}(c_0 + u_0)$. Proposition 3.2 gives (11). Finally the square-root map is even: if u is fixed for R , then $-u$ is fixed for $-R$. Uniqueness proves the last claim. $\square$

The barrier logarithm is therefore exactly

$$L_t = \operatorname{arsinh}(u_0^*), \quad e^{L_t} = \sqrt{1 + (u_0^*)^2} + u_0^*. \quad (12)$$

This connects the fixed point to the logarithmic coordinate used in the prior central-path manuscripts. It does not estimate that coordinate.

5 Global convergence with an explicit error bound

Let $u^{(0)} = 0$ and $u^{(k+1)} = \Phi_{R,t}(u^{(k)})$.

Theorem 5.1 (Deterministic convergence). *For every mask, every $t \geq 1$, and every $k \geq 0$,*

$$\|u^{(k)} - u^*\|_2 \leq Dq^k, \quad D = \sqrt{\frac{nW}{2t}}, \quad q = \sqrt{\frac{2nW}{1+2nW}} < 1. \quad (13)$$

Consequently Picard iteration converges globally from zero. These bounds are exact-arithmetic mathematical bounds; ordinary floating-point computation needs additional roundoff analysis to turn them into interval certificates.

Proof. For a zero-sum vector u , its positive and negative parts have equal total mass. Hence $\|u\|_2^2 \leq \|u\|_1^2/2$. At the fixed point, $\|u^*\|_1 \leq \mathbf{1}^T \sqrt{v + (u^*)^2} = \sqrt{nV}$, which proves $\|u^*\|_2 \leq D$.

Every scalar map $x \mapsto \sqrt{v_j + x^2}$ is one-Lipschitz. Together with $\|R\| \leq 1$, this shows that Φ is nonexpansive, and induction gives $\|u^{(k)} - u^*\| \leq D$. Thus all iterates have norm at most $2D$. On the interval $[-2D, 2D]$ the scalar derivatives have absolute value at most

$$\frac{2D}{\sqrt{t^{-1} + 4D^2}} = q.$$

The line segment joining an iterate and the fixed point stays within these coordinate intervals. The mean-value inequality therefore improves nonexpansiveness to contraction by q for these pairs, proving (13) by induction. $\square$

Corollary 5.2 (A finite-iteration version of the missing estimate). *Let $t = n^2$ and*

$$k_n = \lceil 4(1 + 2n(t + n - 1)) \log n \rceil.$$

Then $\|u^{(k_n)} - u^\|_2 = O(n^{-3/2})$, uniformly in the mask. A proof that $\mathbb{E}(u_0^{(k_n)})^2 = o(1)$ would therefore establish the sufficient root-moment condition in Theorem 6.1 below.*

Proof. Writing $A = 2nW$, one has $\log q = -\frac{1}{2} \log(1 + 1/A) \leq -1/(2(1 + A))$. Thus $q^{k_n} \leq n^{-2}$, while $D = O(\sqrt{n})$. The last assertion follows from the triangle inequality in L^2 . $\square$

The iteration count in this corollary is of order $n^3 \log n$. Proving a moment estimate for two iterations is not a substitute for proving it at this count. The convergence theorem itself is entirely deterministic and supplies no such probabilistic estimate.

6 An exact expectation criterion at the endpoint

The random mask has the same distribution as its complement. All fixed points in a fixed dimension are bounded by Theorem 5.1; in particular the expectations below exist.

Theorem 6.1 (Root-moment criterion). *For every $n \geq 2$ and $t \geq 1$, set*

$$B_{n,t} = \mathbb{E}[\sqrt{1 + (u_0^*(R, t))^2} - 1] \geq 0.$$

Then

$$\sqrt{\frac{t}{W}}(1 + B_{n,t}) \leq \frac{\mathbb{E}\vartheta(G_S)}{\sqrt{n}} \leq \sqrt{\frac{W}{t}}(1 + B_{n,t}). \quad (14)$$

For any sequence $t_n/n \rightarrow \infty$, the MD-02 limit holds if and only if $B_{n,t_n} \rightarrow 0$. Equivalently it holds if and only if

$$\mathbb{E} \min\{(u_0^*(R, t_n))^2, |u_0^*(R, t_n)|\} \rightarrow 0. \quad (15)$$

The simpler condition $\mathbb{E}(u_0^(R, t_n))^2 \rightarrow 0$ is sufficient.*

Proof. Complementation negates u^* , so $\mathbb{E}u_0^* = 0$. Take expectations in (11) to obtain (14). If $t_n/n \rightarrow \infty$, both deterministic prefactors tend to one. Since $B \geq 0$, the two-sided inequality proves both directions of the equivalence.

For every real x ,

$$\frac{1}{1 + \sqrt{2}} \min\{x^2, |x|\} \leq \sqrt{1 + x^2} - 1 \leq \min\{x^2, |x|\}.$$

This follows by rationalizing $\sqrt{1 + x^2} - 1$ and distinguishing $|x| \leq 1$ from $|x| \geq 1$. It proves (15). Finally $\sqrt{1 + x^2} - 1 \leq x^2/2$ proves sufficiency of the second moment. $\square$

Corollary 6.2 (A quantitative sufficient bound). *If $t = n^2$ and $\mathbb{E}(u_0^*)^2 \leq C/n$, then*

$$1 \leq \frac{\mathbb{E}\vartheta(G_S)}{\sqrt{n}} \leq 1 + \frac{C + 1}{2n} + \frac{C}{4n^2}.$$

In particular, a uniform bound $\sup_{k \geq 0} n \mathbb{E}(u_0^{(k)}(R, n^2))^2 \leq C$ would prove MD-02. Neither bound in this corollary is established in this manuscript.

Proof. For the lower bound, pair a graph with its complement and use $\vartheta(G)\vartheta(\bar{G}) = n$ and the arithmetic–geometric mean inequality. For the upper bound apply $\sqrt{1 + y} \leq 1 + y/2$ twice in (14), using $(n - 1)/n^2 \leq 1/n$. For the final implication, convergence of the iterates and Fatou’s lemma transfer a hypothetical uniform bound to u^* . $\square$

7 Uniform moments of the first two Picard roots

This section proves the finite-iterate estimates without assuming independence of the Fourier entries. Let $\mathcal{C}_n$ be the nonzero inversion classes, and $m_c = |c| \in \{1, 2\}$. Define

$$h_c = F^* \text{diag}(1_c) F e_0, \quad r = \sum_{c \in \mathcal{C}_n} \sigma_c h_c.$$

These vectors are real, mutually orthogonal, and satisfy

$$\|h_c\|_2^2 = \frac{m_c}{n}, \quad (h_c)_0 = \frac{m_c}{n}, \quad \mathbf{1}^T h_c = 0. \quad (16)$$

If A is the matrix with columns h_c , then

$$\|A\|_{2 \rightarrow 2}^2 \leq \frac{2}{n}, \quad \|r\|_2^2 = \frac{n-1}{n}, \quad \sum_c m_c^2 \leq 2(n-1). \quad (17)$$

Theorem 7.1 (All-time finite-iterate bounds). *For $a = 1 - t^{-1/2}$, every $n \geq 2$, and every finite $t \geq 1$,*

$$n\mathbb{E}(u_0^{(1)})^2 = a^2 \begin{cases} 2 - 2/n, & n \text{ odd}, \\ 2 - 3/n, & n \text{ even}, \end{cases} \quad (18)$$

$$n\mathbb{E}(u_0^{(2)})^2 \leq 68a^2. \quad (19)$$

Both roots have mean zero. The same conclusions hold at the limit $t \rightarrow \infty$ for these finite iterates, interpreted by continuity.

Proof. Since $\sqrt{v} = t^{-1/2}\mathbf{1} + ae_0$ and $R\mathbf{1} = 0$, $u^{(1)} = ar$. Independence of the class signs gives $\mathbb{E}r_0^2 = n^{-2} \sum_c m_c^2$. For odd n the sum is $2(n-1)$; for even n it is $2(n-2) + 1 = 2n-3$. This proves (18).

Put $b = t^{-1}$ and

$$f(x) = x\sqrt{b + a^2x^2}, \quad g(x) = f(x) - \sqrt{b}x, \quad H = \sum_{j=0}^{n-1} g(r_j).$$

Because $\sum_j r_j = 0$, also $H = \sum_j f(r_j)$. Differentiation yields

$$|g'(x)| \leq 2a|x|, \quad |g''(x)| \leq 2a. \quad (20)$$

For the second bound, when $a, b > 0$ set $z = a|x|/\sqrt{b}$ and use $z(3 + 2z^2)/(1 + z^2)^{3/2} \leq 2$. The cases $a = 0$ and the limit $b \downarrow 0$ follow directly or by continuity.

Flip one class sign σ_c . The corresponding kernel changes from r to $r - 2\sigma_c h_c$. Taylor's theorem and (20) give

$$H(\sigma^{(c)}) - H(\sigma) = -2\sigma_c \langle h_c, g'(r) \rangle + \epsilon_c, \quad |\epsilon_c| \leq 4a\|h_c\|_2^2 = 4am_c/n.$$

Thus, pointwise on the sign cube,

$$\begin{aligned} \sum_c (H(\sigma^{(c)}) - H(\sigma))^2 &\leq 8\|A^T g'(r)\|_2^2 + \frac{32a^2}{n^2} \sum_c m_c^2 \\ &\leq \frac{16}{n} \cdot 4a^2\|r\|_2^2 + \frac{64a^2(n-1)}{n^2} \leq \frac{128a^2}{n}. \end{aligned}$$

The Rademacher Poincare inequality gives

$$\text{Var } H \leq \frac{1}{4} \mathbb{E} \sum_c (H(\sigma^{(c)}) - H(\sigma))^2 \leq \frac{32a^2}{n}. \quad (21)$$

For a self-contained justification of the factor $1/4$, expand a function of independent signs in the orthonormal Walsh basis. Its variance is the sum of squares of all nonconstant coefficients. The right side before the last inequality weights each such coefficient by the size of its nonempty index set, and is therefore no smaller.

The exceptional root weight gives

$$u_0^{(2)} = H + \delta(r_0), \quad \delta(x) = x \left(\sqrt{1 + a^2 x^2} - \sqrt{b + a^2 x^2} \right).$$

The expression in parentheses lies between zero and $1 - \sqrt{b} = a$, so $|\delta(x)| \leq a|x|$. Both H and $\delta(r_0)$ are odd under global sign reversal, so have mean zero. Therefore

$$\mathbb{E}(u_0^{(2)})^2 \leq 2\mathbb{E}H^2 + 2\mathbb{E}\delta(r_0)^2 \leq \frac{64a^2}{n} + \frac{4a^2}{n} = \frac{68a^2}{n}.$$

Continuity in $b \downarrow 0$ and the finite size of the mask space for each fixed n justify the endpoint statement for these two iterates. $\square$

Corollary 7.2 (First-root central limit). *If $t_n \rightarrow \infty$, then $\sqrt{n} u_0^{(1)}(R, t_n)$ converges in distribution to a centered Gaussian of variance 2, along all integer orders.*

Proof. The first root is $a_n \sum_c m_c \sigma_c / \sqrt{n}$. Its variance tends to 2 and the largest summand has magnitude at most $2/\sqrt{n}$. The Lindeberg condition for this independent triangular array is immediate. $\square$

Theorem 7.1 makes no Gaussian approximation and treats the singleton exactly. The constant 68 is a valid explicit bound, not a claimed optimal constant. No central limit theorem for the second or later root is needed for, or asserted by, this theorem.

8 Why the two-iterate theorem is not a solution

There are two distinct limits: $n \rightarrow \infty$ and $k \rightarrow \infty$. Theorem 7.1 bounds the root when $k = 1, 2$ for all barrier times. Theorem 6.1 concerns $k = \infty$. The deterministic error bound in Theorem 5.1 does not bridge this difference at $k = 2$; at $t = n^2$ its bound is of order $\sqrt{n}$ there.

A finite iterate also cannot be silently inserted into (11). The identity $\mathbf{1}^T c = \sqrt{nV}$ used to reconstruct feasible normalized probabilities relies on the fixed-point equation itself. It need not hold for a finite iterate.

Even the smallest deterministic example distinguishes the second iterate from the fixed point. For the empty graph on two vertices, $R = I - P_0$. In the limit $t \rightarrow \infty$,

$$u_0^{(1)} = \frac{1}{2}, \quad u_0^{(2)} = \frac{\sqrt{5} - 1}{4}, \quad u_0^* = \frac{1}{2\sqrt{2}}.$$

The middle and final numbers are unequal. The first two formulas follow directly from the map. For the final formula, the one-variable fixed-point equation is $u = (\sqrt{1 + u^2} - u)/2$, which has the displayed positive solution. This is an illustration of the logical distinction, not a counterexample to the random asymptotic statement.

Likewise a bad deterministic mask at large n cannot by itself refute MD-02: its probability and its contribution to the expectation have to be estimated. The empty and complete masks each have probability $2^{-\lfloor n/2 \rfloor}$. Their combined contribution to the normalized expectation is at most $(n+1)2^{-\lfloor n/2 \rfloor}/\sqrt{n}$, which tends to zero.

The exact outstanding task in this formulation is

$$\mathbb{E}[\sqrt{1 + (u_0^*(R, n^2))^2} - 1] \longrightarrow 0. \quad (22)$$

A bound $\mathbb{E}(u_0^*)^2 = O(1/n)$ would prove substantially more than necessary, but it is not established by the two-iterate argument. The evolving nonlinearity depends on the same random mask as R ; later applications cannot be treated as multiplication by an independent random operator.

9 Reproducibility and claim audit

The accompanying implementation applies R by FFT, computes finite Picard iterates, reports the deterministic exact-arithmetic error formula, and numerically solves the fixed point using a continuation in t . A separate Fourier linear program checks the theta interval. Root-solver residuals are recorded explicitly; these solves are not interval-arithmetic certificates.

The validation script enumerates inversion-class masks at orders 2 through 12 for first- and second-root moments, checks weighted reconstruction and complement symmetry against Fourier linear programs on smaller instances, and records Monte Carlo diagnostics separately. The actual run outcome is in `results/validation_summary.json`. An implemented test or a successful finite experiment is not a proof of (22).

Claim	Status here	Scope
Exact all-time fixed point	Proved	Every mask, every finite $t \geq 1$; Theorem 4.1.
Global Picard convergence	Proved	Explicit deterministic error bound; Theorem 5.1.
Expectation criterion	Proved equivalence	Concerns the converged root; Theorem 6.1.
First two root moments	Proved	All integer orders and all barrier times; Theorem 7.1.
First-root Gaussian limit	Proved	Independent triangular-array limit; Corollary 7.2.
Converged-root moment	Not proved	Equation (22) is the remaining task.
MD-02 expectation limit	Not proved or disproved	No substantive asymptotic subcase of the displayed target is established.
Repository status recommendation	OPEN	Supporting results alone do not establish a displayed-target case.

The mathematical proofs are not accompanied by a claim of independent peer review or historical priority. Earlier proof archives, when available, are preserved as provenance rather than incorporated as unexamined hypotheses. The new theorems above are self-contained.

References

- [1] OpenProblemsInNLA. MD-02, random circulant Lovasz theta, problem statement. <https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/matrix-discrepancy-and-optimization/MD-02>.
- [2] OpenProblemsInNLA. Repository README and contribution guidelines. <https://github.com/ajt60gaibb/OpenProblemsInNLA/blob/main/CONTRIBUTING.md>.
- [3] A. S. Bandeira, J. Blasiok, A. Dmitriev, U. Faure, N. Kireeva, and D. Kunisky. *The Lovasz theta function of random circulant graphs*. arXiv:2502.16227, 2025. <https://arxiv.org/abs/2502.16227>.
- [4] Supplied continuation archive, MD02_continuation_results.zip: *Central-path continuation, sharp initial moments, and certified obstructions*. Sections 2–4 give the Fourier formulation and weighted complementary barriers; subsequent sections give initial-moment results and obstructions.
- [5] Supplied continuation archive, MD02_extended_proofs.zip: *Random circulant theta: initial limit laws and early central-path control*. Sections 2 and 8 give barrier notation and the full-scale gap; Theorem 7.1 gives early-time control.